

Assessment Brief

Module Code	LDS7005M	Module Lecturers		Dr Gayathri Karthick (MD) Haroon Sihan (DMD)	
Module Title	Big Data & Cloud Computing				
Level	7	Credit Value		20	
Assessment Title	(Creative) artefact – A data collection, structuring, and processing of data with an accompanying analytical report				
Workload	N/A				
Assessment Number	1	of	2	Weighting	60%
Submission Type	(Creative) artefact				
Submission Method	Turnitin within Moodle				
Publication Date	18-March-2024				
Due Date	04-June-2024				
Expected Feedback Date	26-June-2024				
Format of Feedback	Written feedback within Turnitin/Moodle				

Programme Learning Outcomes (PLO)

This coursework is designed to achieve the following PLOs:

PLOs **7.1-7.5, 7.8, 7.9**

- 7.1 Critically apply skills, techniques, and knowledge from a range of data analysis methods and algorithms for enhancing and solving problems in various domains.
- 7.2 Develop abstract thinking and design ability to analytically demonstrate concepts relating to data science.
- 7.3 Use research-based knowledge for the design of experiments, analysis, and interpretation of data to provide valid results.
- 7.4 Critically evaluate and analyse advanced data science topics, and concepts, and implement them in workplace.
- 7.5 Identify and implement appropriate programming and software tools to critically analyse big data applications in workplace.
- 7.8 Critically analyse the data and apply predictive modelling technique in the field of Machine Learning and Artificial Intelligence.
- 7.9 Critique legal, social, and ethical issues within the field of data science and applicable ancillary sectors, as applied to contemporary research and industrial practice.

Assignment Description**Assignment Scenario: "Optimizing Big Data Processing in the Cloud with Data Set Extraction and Analysis"**

You are a data architect in a multinational enterprise that handles vast amounts of diverse data across multiple departments. The company is looking to optimize its big data processing capabilities by leveraging cloud computing technologies. Your assignment is to design and implement a solution that utilizes cloud-based services for efficient storage, processing, and analysis of big data. Additionally, include a practical example of data set extraction and analysis. Consider the following aspects in your assignment:

1. **Data Ingestion and Storage (15%):**
 - Describe the types and sources of data that the company deals with.
 - Propose a cloud-based storage solution, considering scalability, redundancy, and cost-effectiveness.
 - Explain the data ingestion process, including tools or services used for seamless data transfer to the cloud.
2. **Scalable Processing Architecture (20%):**
 - Design a scalable big data processing architecture using cloud computing resources.
 - Discuss the choice of cloud services for distributed computing and parallel processing.
 - Explore how the architecture accommodates the company's growing data volumes.
3. **Data Extraction and Pre-processing (15%):**
 - Select a specific data set from the company's domain for extraction and analysis.
 - Discuss the extraction process, including data sources, extraction tools, and any transformations applied.
 - Outline pre-processing steps to clean and prepare the data for analysis.
4. **Data Analysis and Insights (20%):**
 - Implement a cloud-based analytics solution for analysing the selected data set.
 - Provide examples of analytical queries or machine learning algorithms applied to extract meaningful insights.
 - Discuss how the analysis contributes to informed decision-making within the enterprise.
5. **Cost Optimization Strategies (10%):**
 - Develop strategies for optimizing costs associated with storing and processing the selected data set in the cloud.
 - Consider factors such as resource utilization, reserved instances, and cost-effective storage options.
6. **Security and Compliance (15%):**
 - Address security measures to ensure the confidentiality and integrity of the selected data set.
 - Discuss compliance considerations, especially if the enterprise operates in regulated industries.
 - Propose access control mechanisms and encryption practices specific to the analysed data set.
7. **Performance Monitoring and Management (5%):**
 - Outline strategies for monitoring the performance of the big data processing system in the cloud, with a focus on the analysed data set.
 - Discuss how to manage and troubleshoot potential issues or bottlenecks specific to the selected data set.
 - Explore tools or services that enable efficient system management and monitoring.

There are several sources where you can obtain free large datasets for your assignment on big data processing in the cloud. Here are some reputable platforms and repositories:

- Please prioritize working with larger datasets, as your current focus on bigger data sizes and justify this by providing the dataset link in your report.

Eg: Kaggle Datasets, UCI Machine Learning Repository, Google Cloud Public Datasets, AWS Public Datasets, and PhysioNet.

- Please select only one sample topic from your convenience.
 - Healthcare,
 - Chat bot datasets,
 - Healthcare,
 - Augmented/Virtual reality experiences.
 - Reviews using Sentiment analysis.

General considerations

Please be aware that each step should be fully described in your assessment. You should support your implementation with written documentation.

Submit a detailed report addressing each aspect of the assignment, including the specifics of the selected data set and the results of the analysis.

Provide references to relevant cloud computing services, frameworks, or case studies supporting your design decisions.

You should submit a report summarizing your findings (Screenshots with an explanation), including tables and charts to support your analysis. Your report should also include a brief discussion of any limitations or caveats to your design.

Note: Assessment report with a copy of the programme must be converted to pdf file before submitting. Code must be submitted separately as .ipynb file.

Additional Information

The work you present should be your own work, and not just copied from others. You can quote from others, but you must say who the author is and use quotation marks or paraphrase. If you do not do so, we will investigate your work for academic misconduct. This is particularly likely if your Turnitin similarity score is above 25% and/or individual matches are above 6%.

If you require support with your study skills, please visit <https://www.yorksj.ac.uk/students/study-skills/>

Assessment Regulations

Please refer to the York St John University Code of Practice for Assessment and Academic Related Matters 2023-24.

We ask that you pay particular attention to the academic misconduct policy. Penalties will be applied where a student is found guilty of academic and/or ethical misconduct, including termination of programme ([Policy Link](#)).

You are required to keep to the word limit set for an assessment and to note that you may be subject to penalty if you exceed that limit. You are required to provide an accurate word count on the cover sheet for each piece of work you submit ([Policy Link](#)).

For late or non-submission of work by the published deadline or an approved extended deadline, a mark of 0NS will be recorded. Where a re-assessment opportunity exists, a student will normally be permitted only one attempt to be re-assessed for a capped mark ([Policy Link](#)).

An extension to the published deadline may be granted to an individual student if they meet the eligibility criteria of the ([Policy Link](#)).

Please see the assessment criteria below.

York St John University Level 7 Assessment Marking Descriptor- Big Data & Cloud Computing

Criteria	Deliverables	Marks
1. Subject knowledge & understanding	Articulate the types and sources of data while proposing a scalable, redundant, and cost-effective cloud-based storage solution, showcasing expertise in cloud architecture, and detailing the seamless data ingestion process. - Data Ingestion and Storage	15
	Demonstrate profound understanding and proficiency in crafting scalable big data processing architectures in the cloud, emphasizing expertise in selecting cloud services and providing insight into designing solutions for accommodating growing data volumes. - Scalable Processing Architecture	20
	Demonstrate adeptness in selecting a pertinent dataset aligned with project objectives and exhibit a profound understanding of the data extraction process, including source identification, extraction tools, and transformations. Display mastery by outlining pre-processing steps with emphasis on data cleaning and preparation for subsequent analysis. - Data Extraction and Pre-processing	15

2.Higher cognitive skills & originality	Demonstrate advanced cognitive skills through the implementation of a cloud-based analytics solution. Showcase originality by presenting insightful analytical queries or machine learning algorithms and exhibit critical thinking in discussing how the analysis informs decision-making, potentially proposing innovative business strategies. - Data Analysis and Insights	20
	Demonstrate innovative cost optimization strategies, showcasing originality through considerations of resource utilization, reserved instances, and cost-effective storage options in the management of the selected dataset.- Cost Optimization Strategies	5
	Demonstrate addressing security measures, ensuring confidentiality and integrity, and showcasing originality through discussions on compliance, proposing tailored access control mechanisms, and encryption practices for the analysed dataset. - Security and Compliance	10
3. Advanced technical, professional and/or research expertise	Demonstrate advanced technical and professional expertise by outlining strategies for monitoring, management, and troubleshooting in the big data processing system, emphasizing a systematic approach and familiarity with advanced tools in a cloud environment. - Performance Monitoring and Management.	10
4. References and Presentation	Citations are provided in Harvard's style and the report is well organized and presented. Producing a clear and concise written report that summarizes the analysis and findings. Presenting the results in a visually appealing and informative way. Communicating technical concepts in a way that is accessible to a non-technical audience.	5